

David Bedolla, PhD

Robotics Controls | Motion Control & Trajectory Planning | Whole-Body & Mobile Manipulation

Portfolio | [LinkedIn](#) | [GitHub](#) | davidbedollamartinez@outlook.es | Montréal, QC, Canada

PROFESSIONAL PROFILE

Robotics controls researcher and engineer with experience developing and deploying motion-control, trajectory-generation, whole-body control, predictive-control, and robot-manipulation systems on physical hardware. Current work includes a 400 Hz C++/ROS 2 whole-body controller for a 10-DoF mobile manipulator; prior work includes a 1-4 kHz LabVIEW RT/FPGA and C++ control stack for a physical 7-DoF rehabilitation exoskeleton. Background spans robot kinematics and dynamics, real-time control, mobile manipulation, MPC, teleoperation, sensing and actuation, HIL, and hardware validation.

PROFESSIONAL EXPERIENCE

Associate Researcher

Mar 2026-Present | Lab INIT Robots, Montréal

- Develop and iterate a 400 Hz ROS 2/C++ whole-body motion-control pipeline for a physical 10-DoF AgileX and Kinova Gen3 mobile manipulator.
- Implement analytical and Pinocchio Jacobian redundancy resolution for Cartesian tracking, elbow configuration, joint limits, and arm-base coordination; the analytical method executes in approximately 3 microseconds.
- Integrate trajectory execution, world-frame Cartesian control, filtered odometry, teleoperation, command arbitration, validation, and safe state transitions.
- Improved Cartesian trajectory tracking using super-twisting feedback and validated controller behavior on physical hardware.

Postdoctoral Researcher

Mar 2025-Mar 2026 | École de technologie supérieure (ÉTS), Montréal

- Diagnosed a real-time control bottleneck and reduced lower-limb robot latency by 30 times, enabling torque control on target hardware.
- Improved ROS/Python real-time scheduling for autonomous-vehicle control and reduced steering oscillations during testing.
- Developed symbolic dynamics and inverse-kinematics tools for robot modeling, controller development, and validation.
- Delivered 6-DoF inverse-kinematics and real-time motion-control support to international assistive-robotics collaborations.

Research Professor

Mar 2024-Mar 2025 | Universidad Tecnológica de la Mixteca, Mexico

- Developed and tested 7-DoF controller optimization that reduced tracking error by approximately 20 percent and contributed supervision to adaptive-gripper research.

Research Assistant

Jan 2020-Dec 2023 | École de technologie supérieure (ÉTS), Montréal

- Deployed and tuned a 1-4 kHz LabVIEW RT/FPGA and C++ motion-control stack for a physical 7-DoF rehabilitation exoskeleton, including trajectory generation, BLDC actuation, sensing, safety logic, and real-time control.
- Implemented 50 microsecond FPGA PI current regulation, Hall-feedback position tracking, and torque-to-current commands. Characterized friction, deadzones, backlash, and tracking response on hardware.
- Integrated ATI Nano17 force/torque sensing, admittance control, motor-current-based external-force observation, and IMU, EMG, and depth-camera motion acquisition.
- Deployed Gaussian Process residual modeling in robust MPC at approximately 387 microseconds for a 10-step horizon and used predictive uncertainty to adapt robustness margins.
- Achieved approximately 5 microseconds for analytical inverse kinematics, 43 microseconds for learned inverse kinematics, and 210 microseconds for feedforward torque. Reduced symbolic expressions by approximately 90 percent.

Research Professor

Oct 2018-Oct 2019 | UNSIJ, Oaxaca, Mexico

- Implemented real-time neural-network inverse kinematics for a 6-DoF PUMA robot and MATLAB CNN experiments for wildlife detection in camera-trap datasets.

Professor

Feb 2017-Oct 2018 | TecNM, Puebla, Mexico

- Deployed a TI-controller hardware-in-the-loop manipulator emulator at a 10 microsecond sampling period and implemented FPGA voice processing plus computer-vision experiments.

SELECTED ROBOTICS PROJECTS

Whole-Body Mobile Manipulation

Problem: hold a world-frame end-effector target while the base moves. **Contribution:** odometry transforms, arm-base coordination, redundancy handling, and teleoperation. **Hardware:** physical AgileX/Kinova. **Stack:** ROS 2, C++, Pinocchio. **Result:** onboard 400 Hz deployment. [Video](#).

URDF-Driven Trajectory Planning & Motion Generation

Problem: generate reusable joint-space and Cartesian trajectories from robot descriptions. **Contribution:** built a C++/Qt/URDF application with joint-space time laws, Cartesian waypoints, inverse kinematics, B-splines and blended paths, Ruckig motion limits, CAD visualization, and position, velocity, acceleration, and jerk analysis. **Result:** IK tests up to 1 kHz.

ETS-MARSE Real-Time Motion Control & Learning-Based MPC

Problem: constrained exoskeleton tracking with model error and human interaction. **Contribution:** deployed the LabVIEW RT/FPGA and C++ stack, GP-residual robust MPC, force-aware interaction, and human-oriented analytical and learned inverse kinematics on the physical 7-DoF ETS-MARSE. **Result:** 387 microseconds for a 10-step MPC horizon, 5 microseconds for analytical IK, and 43 microseconds for learned IK. [Video](#).

Kinova Gen3 Cartesian Control & Analytical Redundancy

Problem: coordinate Cartesian motion and elbow posture on a redundant physical manipulator. **Contribution:** combined analytical redundancy with Pinocchio Jacobian velocities, joint-limit handling, command arbitration, and PS4 teleoperation. **Result:** approximately 3 microseconds for the analytical redundancy calculation. [Video](#).

Symbolic Robot Dynamics and Real-Time Optimization

Problem: make reusable robot-dynamics expressions fast enough for real-time control. **Contribution:** built custom RNEA tools for Standard and Modified DH models, branched robots, and automatic common-term grouping. **Result:** approximately 90 percent smaller expressions and 210 microseconds for feedforward torque.

TECHNICAL SKILLS

Controls & Planning: Motion control, trajectory generation and planning, Cartesian/task-space control, whole-body control, redundancy resolution, MPC, numerical optimization, torque control, force/admittance control, robust and sliding-mode control

Manipulation & Robot Modeling: Mobile manipulation, robot manipulation, forward/inverse kinematics, Jacobians, rigid-body dynamics, RNEA, teleoperation, HIL, sim-to-real validation

Robotics Software: C++, ROS 2, Python, MATLAB/Simulink, CMake, Qt, Pinocchio, URDF, Ruckig

Real-Time Hardware & Sensing: LabVIEW RT/FPGA, VHDL, deterministic control, BLDC actuation, Hall feedback, DAQ, force/torque sensing, IMU, EMG, depth sensing, digital signal controllers

Robot Learning & Simulation: Gaussian Processes, learning-based MPC, neural-network inverse kinematics, PyTorch reinforcement-learning experiments in simulation, Gazebo, MuJoCo, Isaac Sim

EDUCATION

2020-2023 | PhD in Robotics: Learning-Based Upper-Limb Robotic Rehabilitation, ÉTS, Montréal

2014-2016 | M.Sc. in Robotics: Hardware-in-the-Loop Simulation of a Robotic Manipulator, Universidad Tecnológica de la Mixteca

2009-2014 | B.Eng. in Mechatronics: Universidad Tecnológica de la Mixteca, CENEVAL EGEL Outstanding distinction

Recognition: 2024 ÉTS Excellence Award nomination, 2020 doctoral scholarship, 2014 master's scholarship

SELECTED PUBLICATIONS

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation*. Engineering Applications of Artificial Intelligence, 126, 106966. DOI: [10.1016/j.engappai.2023.106966](https://doi.org/10.1016/j.engappai.2023.106966).

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm*. IEEE PEDS. DOI: [10.1109/PEDS57185.2023.10246735](https://doi.org/10.1109/PEDS57185.2023.10246735).